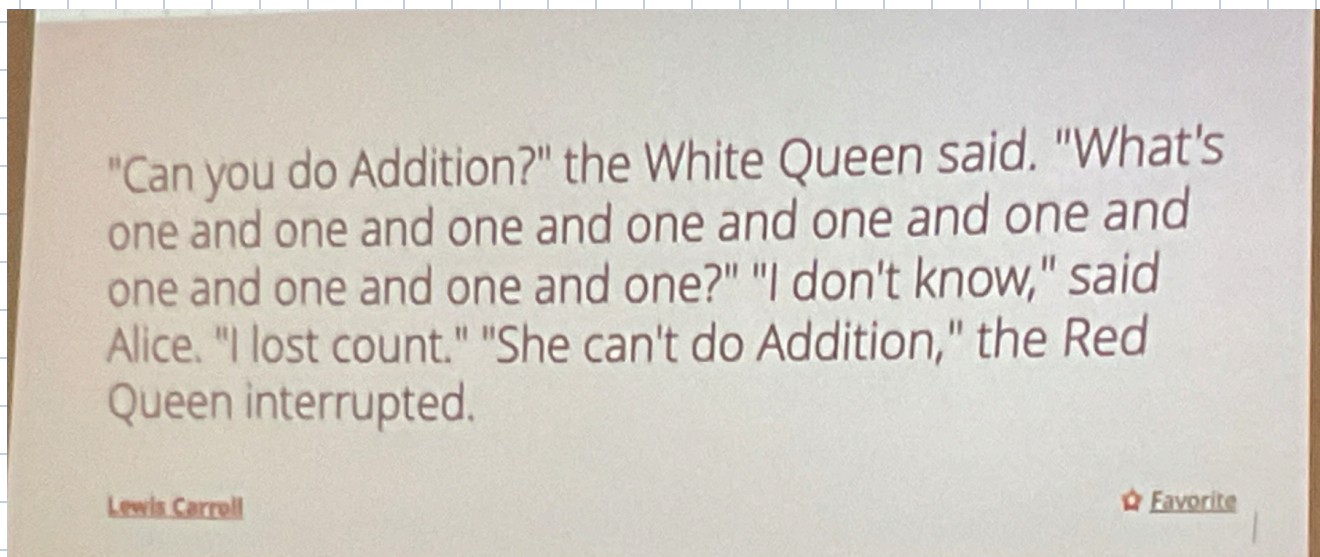


Continuous Charge distribution

+++++
charge rod
line charge

+++++
Sheet of charge
surface charge

Speaker
+++++
Volume charge



Charge Q
+++++

 $L = 1.5 \text{ m}$

linear charge density
 $\frac{Q}{L} = \lambda$

$$Q = 1.5 \lambda$$

$$dQ = \lambda dx$$

A
Q

Surface charge density

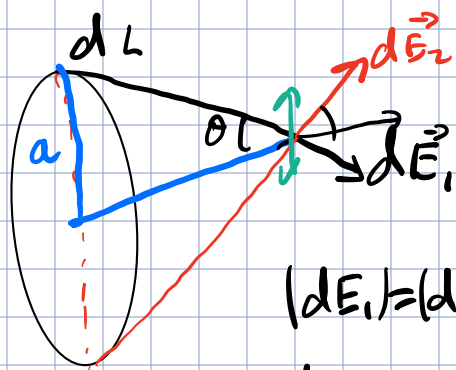
$$\sigma = \frac{Q}{A}$$

volume charge density
Q

P

$$P = \frac{Q}{V}$$

E - field of a charged ring



$$|dE_1| = |dE_2| = dE$$

$$dE_1 \cos \theta + dE_2 \cos \theta$$

$$\frac{Q}{2\pi a} = \lambda$$

$$dQ = \lambda dL$$

$$E = \int 2 dE \cos \theta = \int \frac{2 k \lambda dL}{x^2 + a^2} \cdot \frac{x}{(x^2 + a^2)^{3/2}} = \frac{2k\lambda x}{(x^2 + a^2)^{3/2}} dL$$

$$E = \frac{2k\lambda x}{(x^2 + a^2)^{3/2}} \int dL = \frac{2k\lambda x \pi a}{(x^2 + a^2)^{3/2}} = Q$$

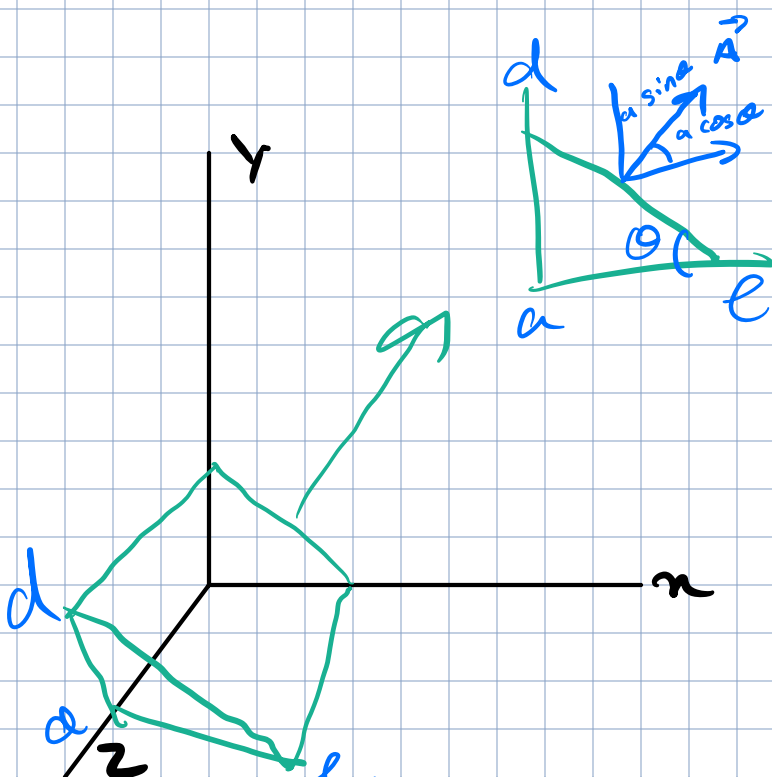
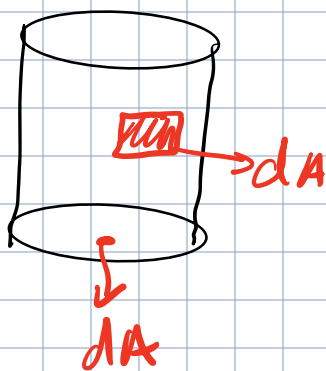
$$E = \frac{kQx}{(x^2 + a^2)^{3/2}}$$

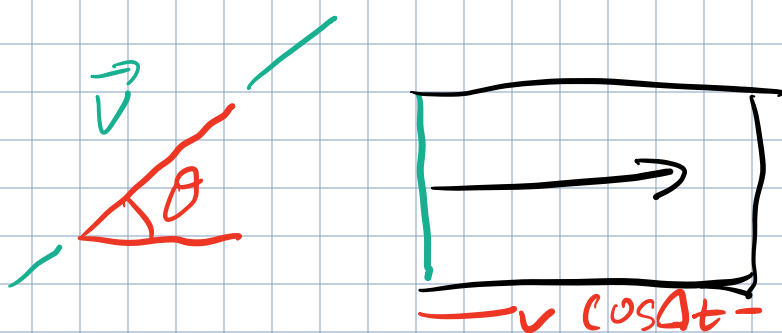
for $x \gg a$

$$E = \frac{kQx}{x^3} = \frac{kQ}{x^2}$$

Vector Area

Area faces you





$$\Delta V = A V \cos \theta \Delta t$$

↑
volume

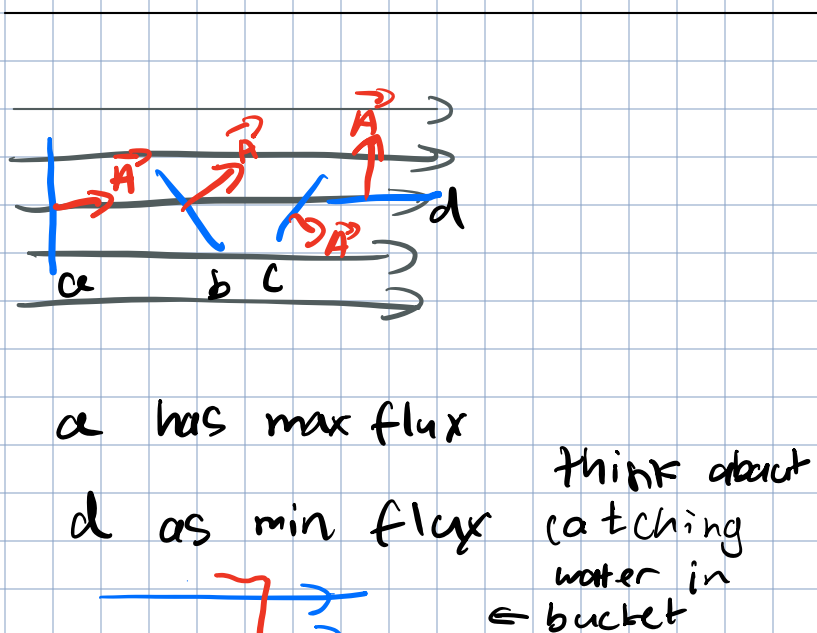
$$\frac{\Delta V}{\Delta t} = A V \cos \theta$$

$$\vec{V} \cdot \vec{A} \rightarrow \text{gallons/m}^2$$

flux

Electrical flux

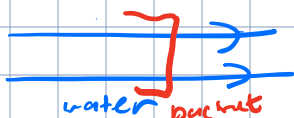
$$\Phi = \vec{E} \cdot \vec{A} = E A \cos \theta$$



a has max flux

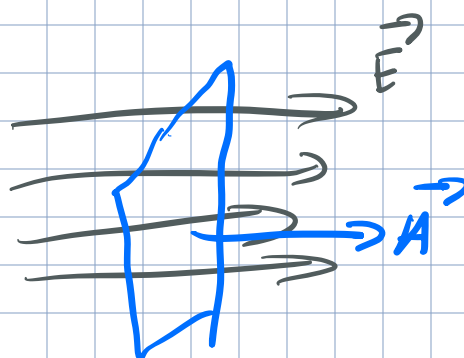
d as min flux

think about
catching
water in
bucket



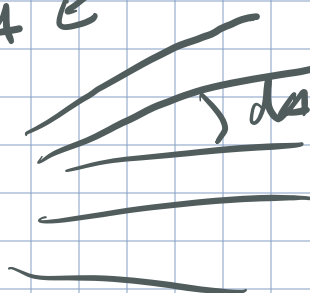
* only area in the direction of

E field matters not area of face *

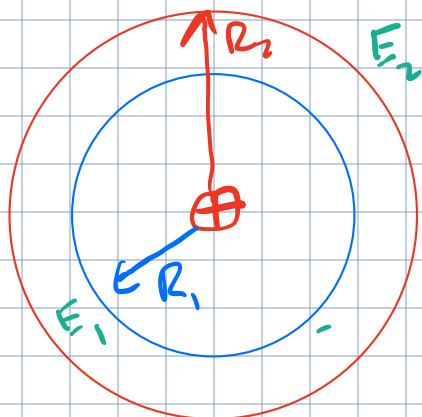
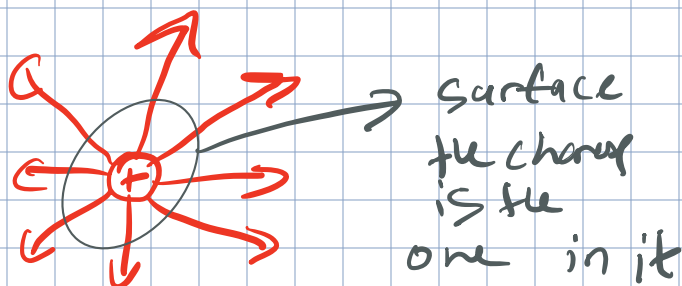


if not same

$$\int dA$$



Closed gaussian surface



$$E_1 = \frac{kQ}{R_1^2}$$

$$k = \frac{1}{4\pi\epsilon_0}$$

$$E_1 \cdot 4\pi R_1^2 = \frac{kQ}{R_1^2} \cdot 4\pi R_1^2$$

$$= 4\pi kQ = Q\epsilon_0$$

So radius doesn't matter

Gauss's law

$$\Phi = \int \vec{E} \cdot d\vec{A} = \frac{\text{charge}}{\epsilon_0}$$

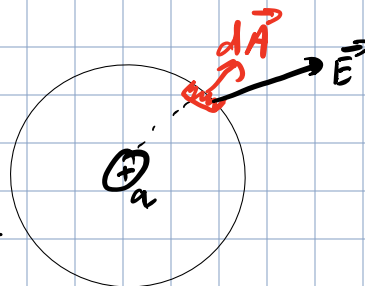
Day 2

ex1 $\Phi = \int_{\text{GS}} \vec{E} \cdot d\vec{A} = \int E dA$

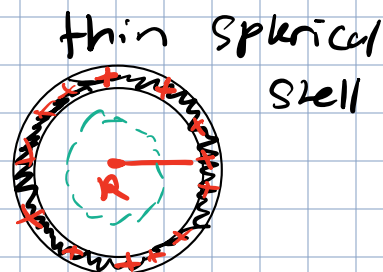
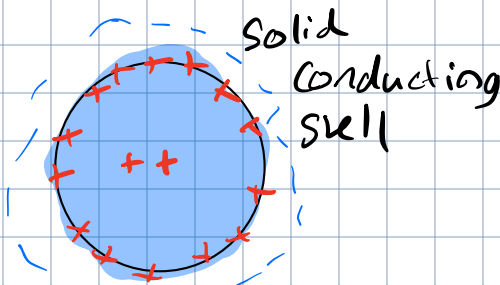
$$= E \int da$$

$$= 4\pi r^2 E$$

Since E will be constant over a sphere



Examples



$$\int_{r < R} \vec{E} \cdot d\vec{A} = \frac{q_{enc}}{\epsilon_0} = 0 \quad q_{enc} = 0 \quad \text{no charge in middle}$$

$$E(r < R) = 0$$

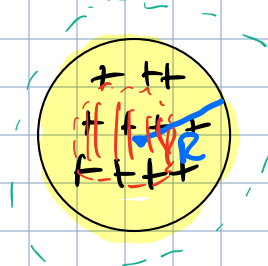
$r > R \rightarrow$ for field outside charge

$$\int \vec{E} \cdot d\vec{A} = 4\pi r^2 E$$

$$= \frac{q_{enc}}{\epsilon_0} = \frac{Q}{\epsilon_0}$$

$$E(r > R) = \frac{Q}{4\pi r^2 \epsilon_0} = \frac{kQ}{r^2}$$

Insulator Sphere



$r < R$

$$\rho = \frac{Q}{\frac{4}{3}\pi R^3}$$

vol charge density

$r > R$

$$\int \vec{E} \cdot d\vec{A} = \frac{q_{enc}}{\epsilon_0} = \frac{Q}{\epsilon_0}$$

$$E 4\pi r^2 = \frac{Q}{\epsilon_0}$$

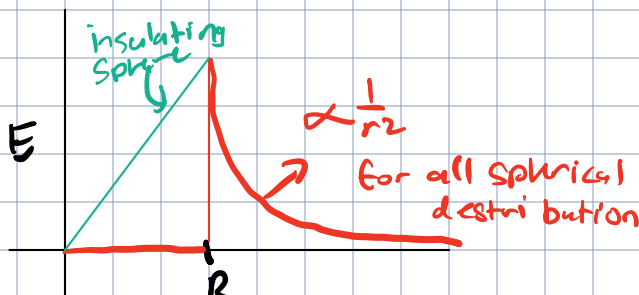
$$E = \frac{Q}{4\pi \epsilon_0 r^2} = \frac{kQ}{r^2}$$

$$\int_{G_S} \vec{E} \cdot d\vec{A} = E 4\pi r^2$$

$$= \frac{\frac{4}{3}\pi r^3 \rho}{\epsilon_0} = \frac{\frac{4}{3}\pi r^2 Q}{\frac{4}{3}\pi R^3 \epsilon_0} = \frac{Q}{\epsilon_0} \frac{r^3}{R^3}$$

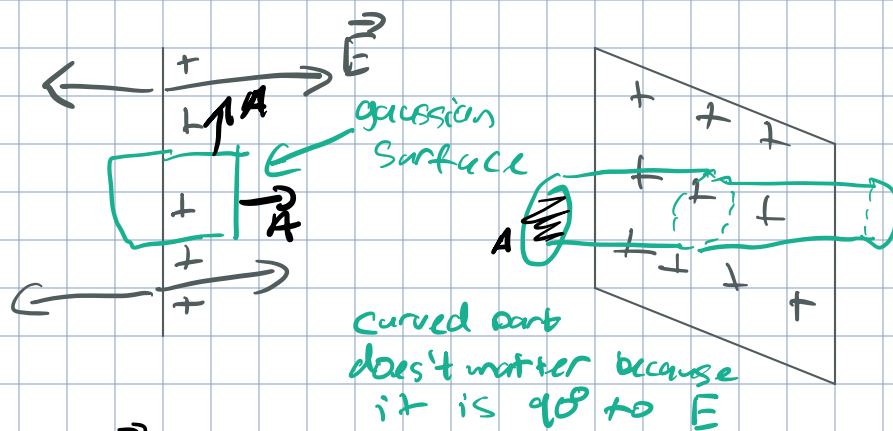
$$4\pi r^2 E = \frac{Q}{\epsilon_0} \frac{r^2}{R^3} \rightarrow E = \frac{Q}{4\pi \epsilon_0 R^3} r$$

Summary



Shells + solid
conduction sphere

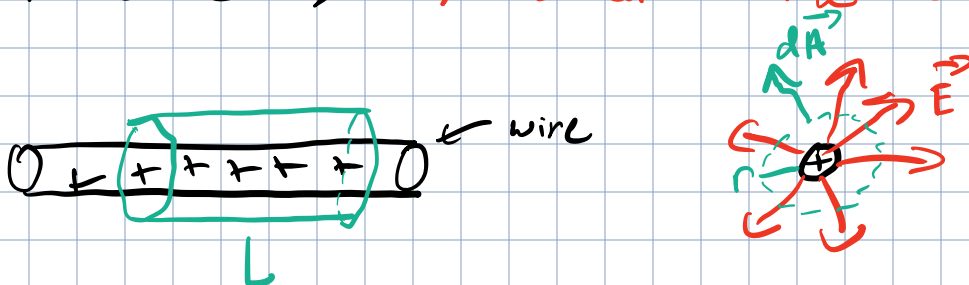
Infinite (large) sheet of charge



$$\int_{GS} \vec{E} \cdot d\vec{A} = 2EA = \frac{q_{enc}}{\epsilon_0} = \frac{\sigma A}{\epsilon_0}$$

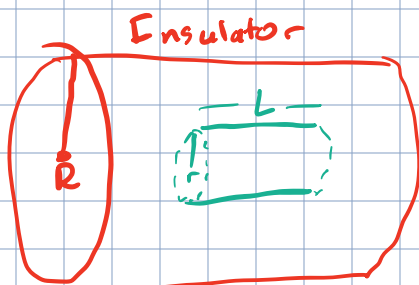
$$E = \frac{\sigma}{2\epsilon_0}$$

Line of charge $\lambda = \text{linear charge density}$



$$\int \vec{E} \cdot d\vec{A} = E 2\pi r L = \frac{q_{enc}}{\epsilon_0} = \frac{\lambda L}{\epsilon_0}$$

$$E = \frac{\lambda}{2\pi\epsilon_0 r}$$



$$P = \frac{Q}{V}$$

$$\int \vec{E} \cdot d\vec{A} = E 2\pi r L = \frac{q_{enc}}{\epsilon_0} = \frac{2\pi r L P}{\epsilon_0}$$

$$E = \frac{P r}{2\epsilon_0}$$